

LM06 Simulation Methods

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Version 1.0

1. Introduction

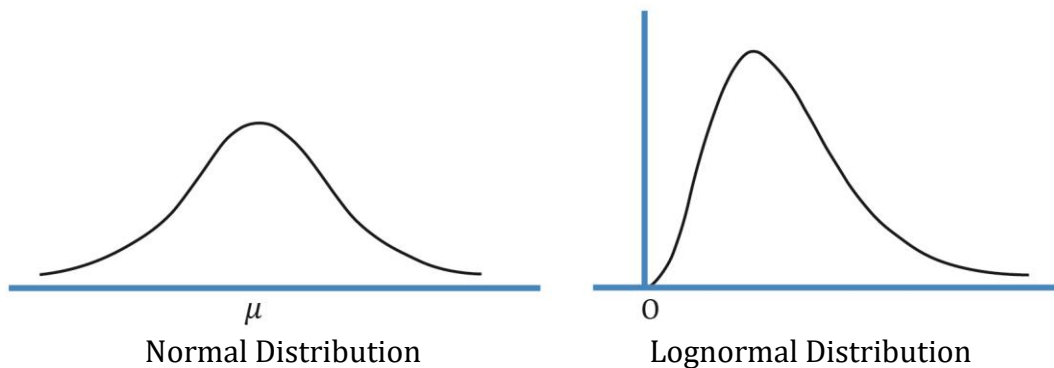
This learning module covers:

- Lognormal distribution and continuously compounded asset return
- Monte Carlo simulation
- Bootstrapping

2. Lognormal Distribution and Continuous Compounding

The Lognormal Distribution

If x is a random variable that is normally distributed, then to create a lognormal distribution of x we take e^x and plot the values on a graph.



Instructor's Note: It is called a 'lognormal' distribution because 'the log is normal'.

The properties of a lognormal distribution are:

- It cannot be negative. It is bounded to the left by '0'.
- The upper end of its range extends to infinity.
- It is skewed to the right i.e. it has a long right tail.

Both normal and lognormal distributions are important for finance professionals.

A normal distribution is typically used to model asset returns, because returns generally vary about the mean, with a high probability of returns being close to the mean. Returns can also sometimes be negative.

However, asset prices do not vary equally about a mean price, since the probability of extreme changes in price decreases as the price approaches zero. This means asset prices will not form a symmetrical graph like that of the normal distribution. Instead, asset prices follow a lognormal distribution, which is skewed to the right and cannot be negative.

Continuously Compounded Rates of Return

The continuously compounded rate of return can be calculated as the natural logarithm of the ending price over the beginning price.

$$R_c = \ln (P_t/P_0)$$

A property of the continuously compounded rate of return is that they are additive. The continuously compounded return from period 0 to period T is the sum of the incremental one-period returns between 0 and T.

$$r_{0,T} = r_{T-1,T} + r_{T-2,T-1} + \dots + r_{0,1}$$

This is illustrated in the below example.

Example:

(This is based on Practice Problem 1 from the curriculum.)

The weekly closing prices of a share are as follows:

Date	Closing Price
1 August	112
8 August	160
15 August	120

Calculate the continuously compounded return of this share for the period August 1 to August 15.

Solution:

The continuously compounded return of an asset over a period is equal to the natural log of the asset's price change during the period. In this case, $\ln(120/112) = 6.90\%$.

The continuously compounded return from period 0 to period T is also equal to the sum of the incremental one-period continuously compounded returns, which in this case are weekly returns.

Week 1 return: $\ln(160/112) = 35.67\%$.

Week 2 return: $\ln(120/160) = -28.77\%$.

Continuously compounded return = $35.67\% + -28.77\% = 6.90\%$

Continuously compounded returns are used in many asset pricing models, as well as in risk management.

Many investment applications make the assumption that returns are **independently and identically distributed (i.i.d.)**. The concept of independence captures the idea that investors cannot predict future returns based on past returns. The concept of identical distribution captures the assumption of stationarity, which implies that the mean and variance of return do not change from period to period.

Calculating Volatility

The volatility (measured in terms of standard deviation) of the continuously compounded returns on an asset is typically stated as an annualized number. However, in practice volatility is often estimated using a historical series of continuously compounded daily

return. To convert daily volatility into annual volatility, we annualize based on 250 days in a year (the approximate number of business days that financial markets are open for trading). For example, if daily volatility is 0.01, the annual volatility will be $0.01\sqrt{250} = 0.1581$

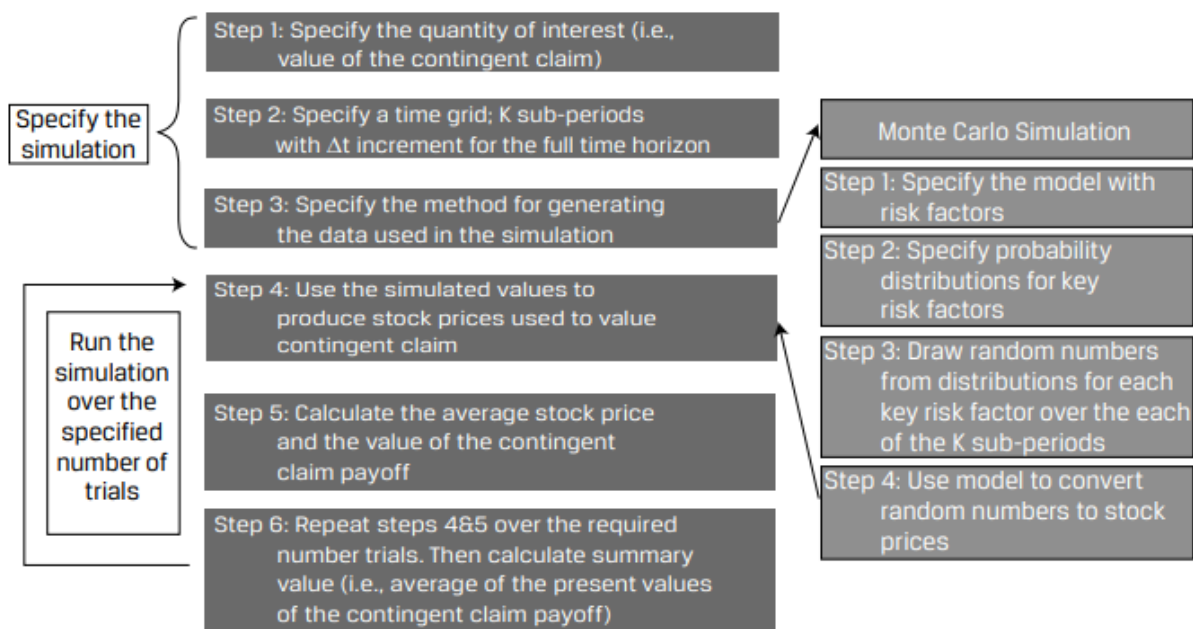
3. Monte Carlo Simulation

Monte Carlo simulation is a computer simulation used to simulate possible security prices based on risk factors. As input, it uses randomly generated values for risk factors based on their assumed probability distributions. It processes this information as per the specified model and runs thousands of iterations. Then it gives the distribution of the expected value of the security as output.

Monte Carlo simulation is widely used to estimate risk and return in investment applications. It is also used as a tool for valuing complex securities with embedded options where no analytic pricing formula is available (e.g. Mortgage backed securities).

A limitation of Monte Carlos simulation is that it is fairly complex and will provide answers that are no better than the assumptions. Also, simulation is not an analytical method but a statistical one. It cannot provide more insight into cause-and-effect relationships like analytical methods.

Exhibit 5 from the curriculum shows the steps for implementing a Monte Carlo simulation for valuing a contingent claim (an option).



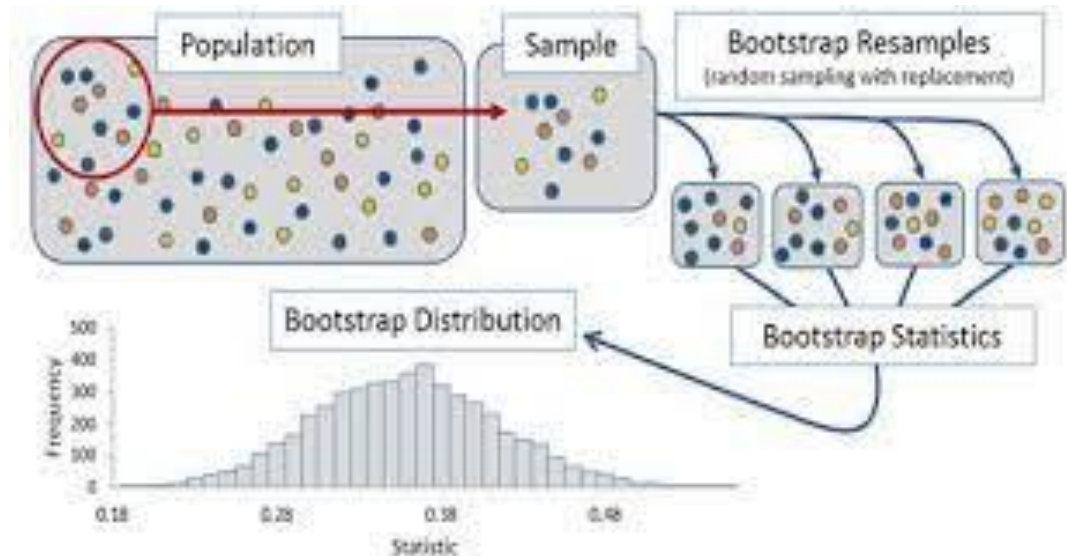
4. Bootstrapping

In an earlier lesson, we learnt how to find the standard error of the sample mean based on the central limit theorem. We will now cover 'resampling': a computational tool in which we

repeatedly draw samples from the original observed data sample for the statistical inference of population parameters. A popular resampling method is bootstrap.

Bootstrap

The bootstrap technique is illustrated in the figure below:



The technique is used when we do not know what the actual population looks like. We simply have a sample of size n drawn from the population. Since the random sample is a good representation of the population, we can simulate sampling from the population by sampling from the observed sample i.e., we treat the randomly drawn sample as if it were the actual population.

Under this technique, samples are constructed by drawing observations from the large sample (of size n) one at a time and returning them to the data sample after they have been chosen. This allows a given observation to be included in a given small sample more than once. This sampling approach is called sampling with replacement.

If we want to calculate the standard error of the sample mean, we take many resamples and calculate the mean of each resample. We then construct a sampling distribution with these resamples. The bootstrap sampling distribution will approximate the true sampling distribution and can be used to estimate the standard error of the sample mean. Similarly, the bootstrap technique can also be used to construct confidence intervals for the statistic or to find other population parameters, such as the median.

Bootstrap is a simple but powerful technique that is particularly useful when no analytical formula is available to estimate the distribution of estimators.

Summary

LO: Explain the relationship between normal and lognormal distributions and why the lognormal distribution is used to model asset prices when using continuously compounded asset returns.

If x is a random variable that is normally distributed, then to create a lognormal distribution of x we take e^x and plot the values on a graph.

A lognormal distribution is often used to model asset prices because the asset prices need to be positive, they cannot be negative.

LO: Describe Monte Carlo simulation and explain how it can be used in investment applications.

Monte Carlo simulation is a computer simulation that produces a distribution of possible security prices using randomly generated values for risk factors, based on their assumed distributions.

Monte Carlo simulation is widely used to estimate risk and return in investment applications. It is also used as a tool for valuing complex securities with embedded options where no analytic pricing formula is available.

Limitations include:

- It is fairly complex and will provide answers that are no better than the assumptions.
- Simulation is not an analytical method but a statistical one.

LO: Describe the use of bootstrap resampling in conducting a simulation based on observed data in investment applications.

Bootstrap treats a randomly drawn sample as if it were the population. It constructs the sampling distribution of an estimator by repeatedly drawing samples from the original sample with replacement.